

2 The nervous system of an organism enables it to respond to a stimulus.

(a) A reflex action is a rapid involuntary movement in response to a stimulus.
The response to a pinprick in a finger is an example of a reflex action.

(i) Which component of the nervous system continues a reflex arc immediately after the receptor has been stimulated by the stimulus?

(1)

- ☐ **A** motor neurone
- ☐ **B** relay neurone
- ☐ **C** Schwann cell
- ☐ **D** sensory neurone

(ii) Which pathway shows a reflex arc?

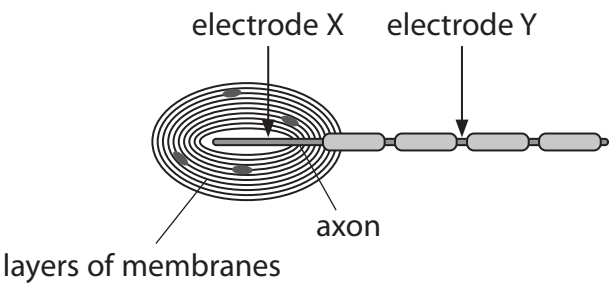
(1)

- ☐ **A** muscle —————> receptor —————> brain
- ☐ **B** muscle —————> spinal cord —————> brain
- ☐ **C** receptor —————> spinal cord —————> muscle
- ☐ **D** spinal cord —————> brain —————> muscle

(b) Pacinian corpuscles are pressure receptors found in the skin of the fingertip.

The effect of different pressures applied to the finger tip, on electrode potentials across the axon membrane of the neurone of a Pacinian corpuscle was investigated.

The diagram shows the structure of a Pacinian corpuscle and the location of electrodes, X and Y, used to measure axon membrane potentials.



The table shows the results of the investigation.

Pressure applied to the fingertip	Membrane potential at electrode X / mV	Membrane potential at electrode Y / mV
None	-70	-70
Low	-60	-70
Medium	+20	+40
High	+40	+40

(i) Describe how the resting potential of -70mV is maintained in the axon when no pressure is applied.

(3)

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(ii) Suggest how a change in pressure at the Pacinian corpuscle causes an action potential in the axon.

(3)

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(iii) Explain why the membrane potential at electrode Y was the same when medium or high pressure was applied to the fingertip.

(3)

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(Total for Question 2 = 11 marks)